

22 Compounds **P** to **S** have the same molecular formula C_7H_7X , where X can be Cl, Br or I.

Each compound was separately heated with ethanolic $AgNO_3$ and the observations were recorded.

compound	observations
P	precipitate formed after 2 minutes
Q	no precipitate formed
R	precipitate formed instantaneously
S	precipitate formed after 10 minutes

Which statement about the compounds **P** to **S** is correct?

- A** The C—X bond length increases in the order **R** < **P** < **S** < **Q**.
- B** **R** can be formed by reacting Cl_2 with methylbenzene in the presence of UV light.
- C** **Q** can be formed by reacting Br_2 with methylbenzene in the presence of $AlBr_3$.
- D** Heating of each compound **P**, **Q**, **R** and **S** in acidified $KMnO_4$ produces $C_6H_5CO_2H$.